

Ramiro Barcala Roca

Junior Software Developer / Systems Technician

📍 Flores, CABA, Argentina

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ABOUT ME

Systems Analysis and Computer Engineering student at FIUBA passionate about low-level systems programming, distributed architectures, and computer organization. I combine a strong technical foundation in electronics with analytical problem-solving skills to build robust, concurrent, and high-performance software.

EDUCATION

B.S. in Systems Analysis & Computer Engineering

2023 — Present

Faculty of Engineering, University of Buenos Aires (FIUBA) — Dual Degree Program

Core courses: Algorithms & Data Structures, C++ Programming Workshop, Computer Organization, Computer Networks, Data Management.

Electronics Technician

2017 — 2022

Escuela Cristiana Evangélica Argentina (ECEA)

6-year technical degree: PCB modeling/fabrication, digital logic, microcontrollers, and precision lab instrumentation.

WORK EXPERIENCE

Repair & Hardware Technician

2022 — 2023 (1 year)

Sitek Electrónica SRL

- **Hardware Diagnosis:** Systematic fault detection and repair in industrial microelectronic components.
- **Industrial Maintenance:** Reconditioning of PLC power supply modules and execution of preventive maintenance plans.
- **Circuit Design:** Modeling, simulation, and manufacturing of custom printed circuit boards (PCBs).

FEATURED PROJECTS

Argentum Online — Client/Server Engine

C++17 • TCP Sockets • Multithreading • SDL2

Engineered a multiplayer client-server online game replica. Designed non-blocking TCP socket layers, custom binary protocol serialization, multithreaded synchronization using mutexes, and design patterns (MVC, State, Observer).

Data Research & Predictive Modeling

Python • Pandas • Scikit-Learn • EDA

Exploratory Data Analysis (EDA) pipelines and machine learning predictive models across weather, music streaming, and urban transit datasets.

Network Protocol Analysis & Diagnostics (DNS & ICMP)

Linux • Wireshark • DNS • Bash • LaTeX

Deep packet inspection and analysis of iterative vs. recursive name resolution using `dig` and Wireshark. Authored formal technical documentation in LaTeX.

LANGUAGES & SYSTEMS

Programming: C, C++, Rust, Python, Java, Assembler, Clojure, Smalltalk.

Systems & Networks: Linux (Ubuntu/Debian), TCP/IP, Sockets, Wireshark, Bash, Git.

HARDWARE & LANGUAGES

Hardware: PCB Design/Routing, PLCs, AutoCAD, FreeCAD, LaTeX.

Languages: Spanish (Native), English Advanced B2 (FCE Oxford College, Grade B).